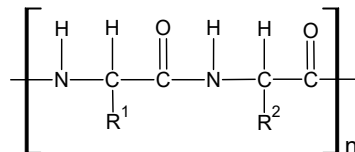


- 27 The diagram below shows the general structure of a protein.



Chymotrypsin is an enzyme that hydrolyses protein into smaller peptides and amino acids. It specifically hydrolyses the peptide bond on the carboxylic end of phenylalanine (Phe).

The structure of hexapeptide Y and the M_r of selected amino acids are given below.

Hexapeptide Y: Val–Ala–Lys–Phe–Ser–Arg

Amino acid	M_r
Valine (Val)	117
Alanine (Ala)	89
Lysine (Lys)	146
Phenylalanine (Phe)	165
Ser (Serine)	105
Arginine (Arg)	174

What are the M_r of the two fragments obtained when hexapeptide Y is hydrolysed by chymotrypsin?

	M_r of fragment 1	M_r of fragment 2
A	517	279
B	463	261
C	316	408
D	352	444